

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1104

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6

Total Marks:60

Annexure A

Application Based Questions

Code of Conduct:

*I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Judy Allen J

1. Use Case Diagram Analysis

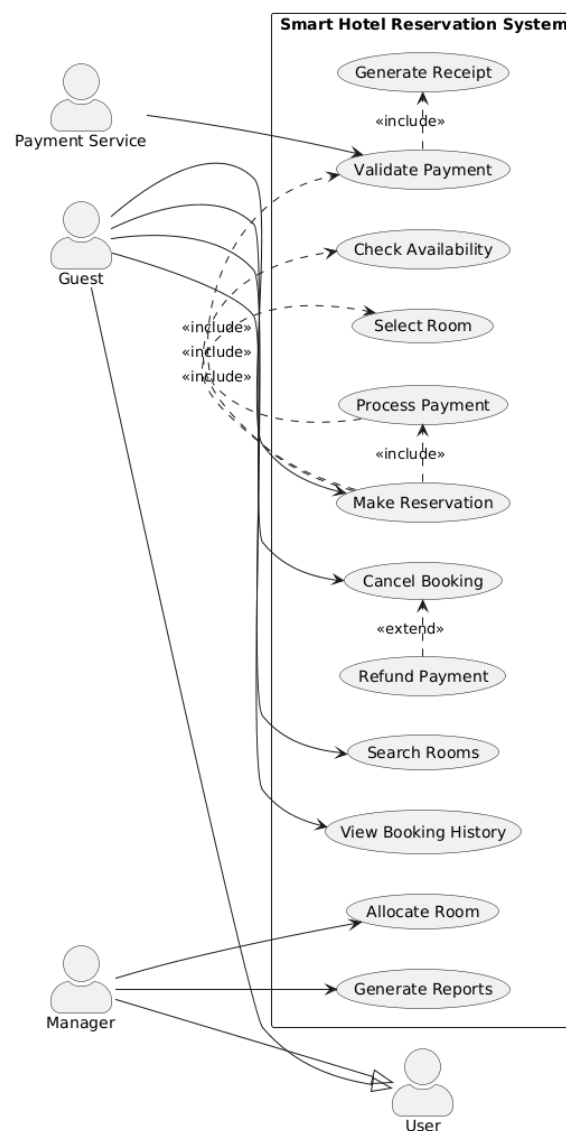
A Smart Hotel Reservation System allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.

Analyze the system requirements, identify missing actors and use cases, and design an optimized Use Case Diagram showing proper relationships (include, extend, generalization) for better system clarity.

Problem Analysis

The Smart Hotel Reservation System allows guests to search rooms, make reservations, cancel bookings, and complete payments. Managers allocate rooms and generate reports, while the Payment Service validates transactions. The redesigned diagram adds missing actors, use cases, and UML relationships for better clarity and functionality.

Use Case Diagram



Explanation

- Guest searches rooms, books, cancels bookings, and views booking history.
- Manager allocates rooms and generates reports.
- Payment Service validates payments.
- <<include>> is used for mandatory functions like payment and availability checking.
- <<extend>> is used for refund during booking cancellation.

Analysis

- Clearly separates actor responsibilities.
- Uses correct UML relationships.
- Improves system organization and maintainability.
- Covers all required system functions.

Conclusion

The optimized Use Case Diagram accurately represents the hotel reservation system using proper actors, use cases, and UML relationships, resulting in a clear and scalable system design.

2. Class Diagram Design

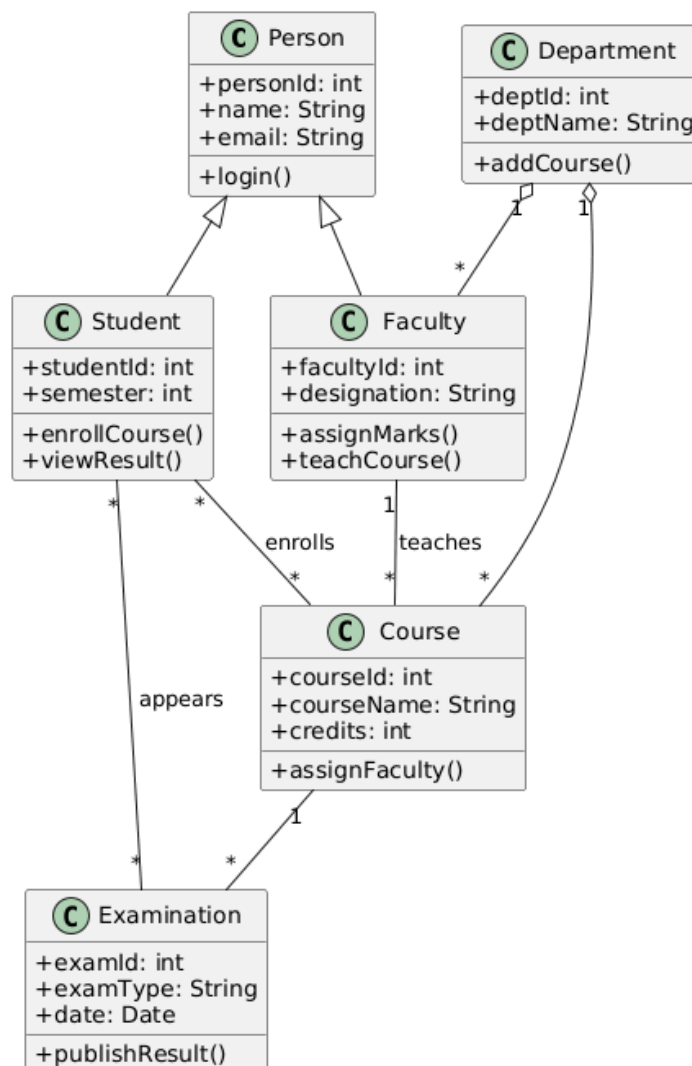
A University Management System includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.

Analyze the system, identify relevant classes with attributes and methods, define relationships (association, aggregation, inheritance), and construct a well-structured Class Diagram.

Problem Analysis

The University Management System manages students, faculty, courses, departments, and examinations. The existing design lacks proper attributes, methods, and relationships. The optimized Class Diagram defines these elements using association, aggregation, and inheritance to improve system organization.

Class Diagram



Explanation

- Person is the parent class of Student and Faculty.
- Department contains multiple courses and faculty members.
- Students enroll in courses.
- Faculty teach courses.
- Examinations are conducted for courses and attended by students.

Analysis

- Uses correct inheritance, association, and aggregation.
- Clearly defines class responsibilities.
- Improves modularity and maintainability.
- Represents the university system accurately.

Conclusion

The optimized Class Diagram effectively models the University Management System with proper classes, attributes, operations, and relationships, resulting in a clear and scalable object-oriented design.

3. Sequence Diagram Evaluation

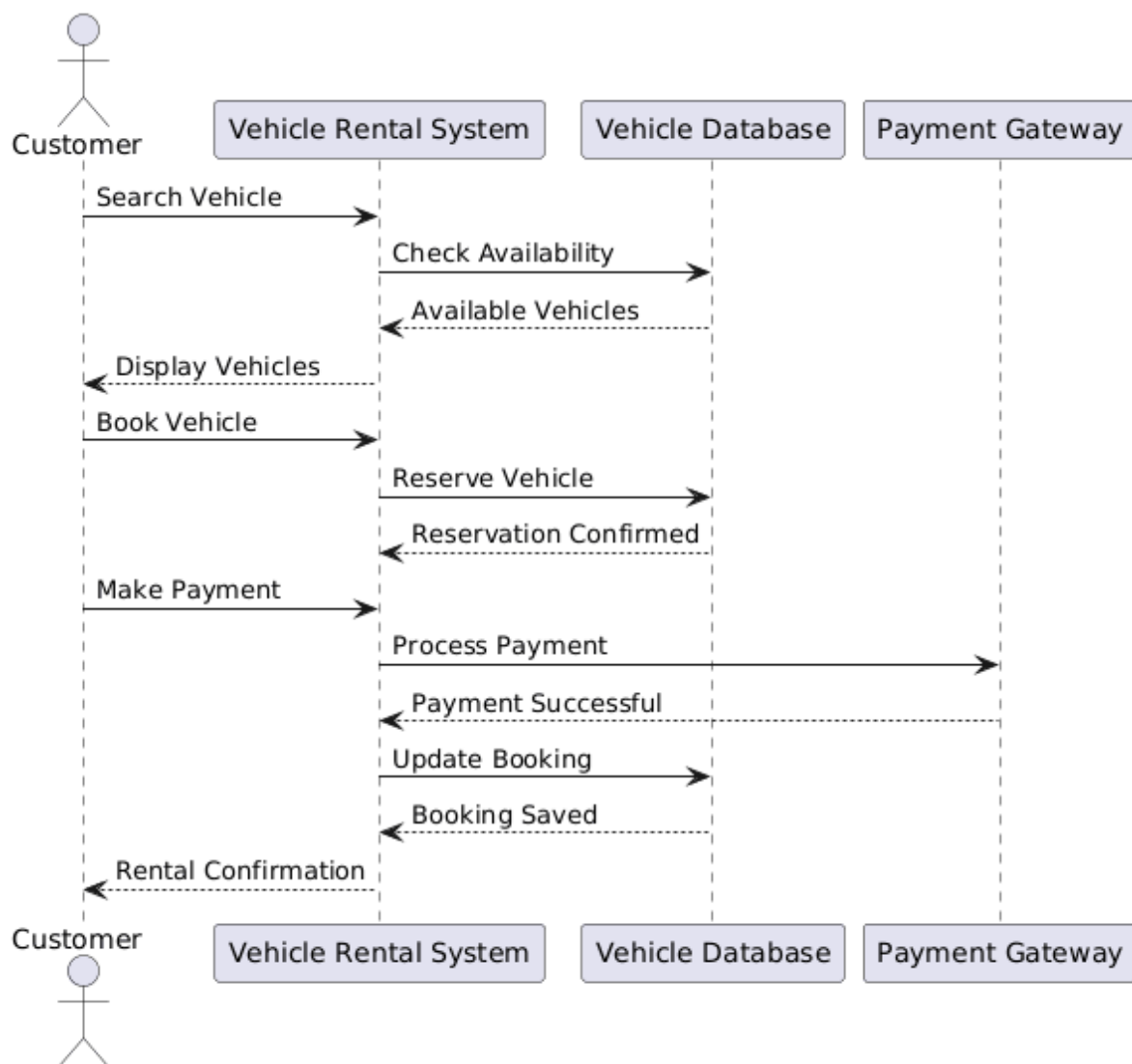
In a Vehicle Rental System, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.

Analyze the process flow, identify missing objects and message sequences, and design a detailed Sequence Diagram representing proper interaction among system components.

Problem Analysis

The Vehicle Rental System allows customers to search for vehicles, book a vehicle, make payments, and receive rental confirmation. The optimized Sequence Diagram clearly defines the interaction between users, the system, payment service, and database for an efficient rental process.

Sequence Diagram



Explanation

- The Customer searches and books a vehicle.
- The System checks vehicle availability and reserves the selected vehicle.
- The Payment Gateway processes the payment.
- The Database stores the booking details.
- Finally, the system sends the rental confirmation to the customer.

Analysis

- Shows the correct sequence of interactions.
- Clearly represents communication between all system components.
- Ensures accurate booking and payment processing.
- Improves system clarity and workflow.

Conclusion

The Sequence Diagram accurately illustrates the interaction between the customer, system, database, and payment gateway, providing a clear and organized workflow for the Vehicle Rental System.

4. State Diagram Construction

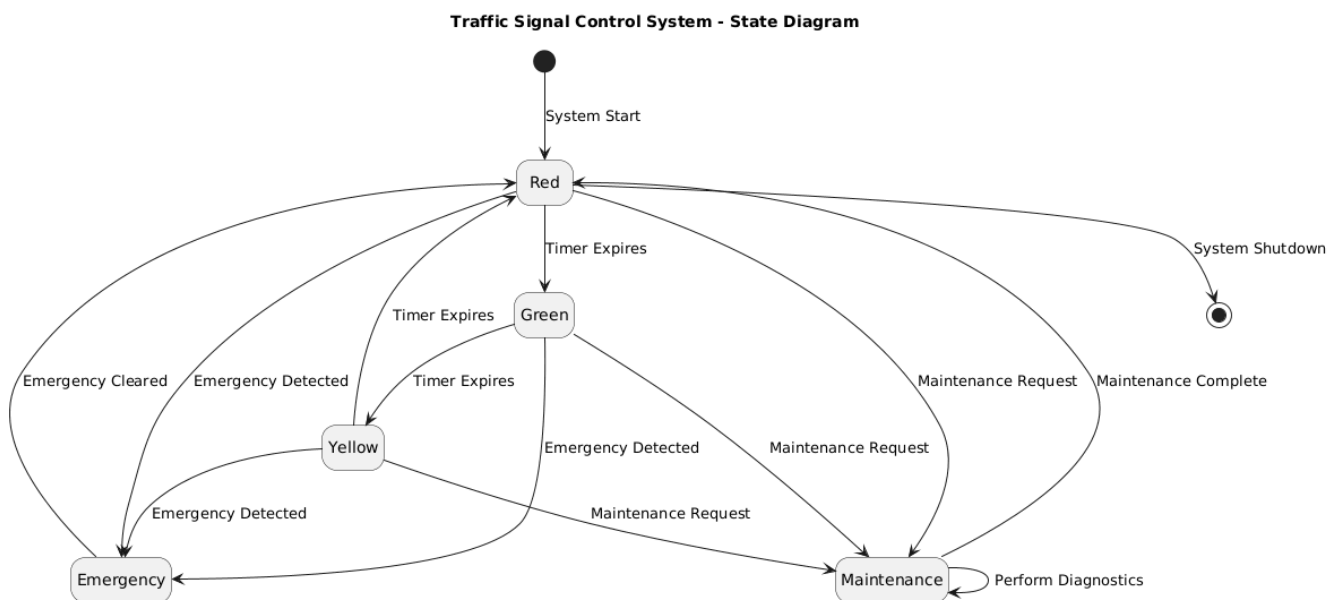
A Traffic Signal Control System operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.

Analyze the system behavior, identify valid states and transitions, and construct a State Transition Diagram ensuring all events and conditions are properly represented.

Problem Analysis

The Traffic Signal Control System operates through different signal states such as Red, Green, Yellow, Emergency, and Maintenance. The optimized State Diagram defines the valid state transitions based on events, ensuring safe and efficient traffic control.

State Diagram



Explanation

- The signal starts in the Red state.
- It changes from Red → Green → Yellow → Red based on timer events.
- During emergencies, the system enters Emergency Mode.
- Maintenance Mode is activated during servicing.
- After emergency or maintenance, the signal returns to Red. \

Analysis

- Clearly defines all states and transitions.
- Handles emergency and maintenance conditions.

- Ensures smooth and safe traffic signal operation.
- Improves system reliability.

Conclusion

The State Diagram accurately represents the behavior of the Traffic Signal Control System by defining all major states, events, and transitions, resulting in a reliable and well-structured design.

5. Component Diagram Design

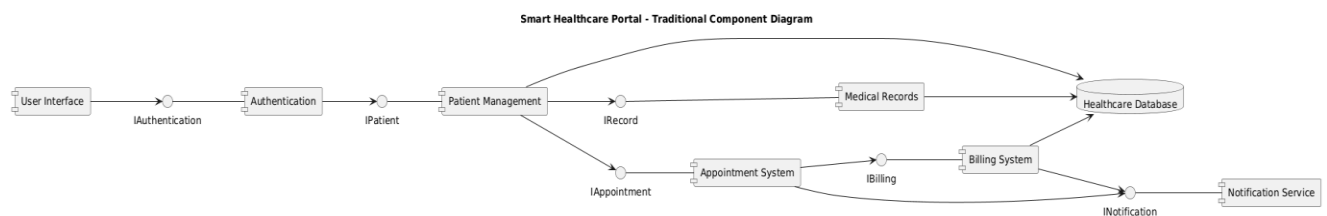
A Smart Healthcare Portal consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.

Analyze the system architecture, identify major components and their interactions, and design a Component Diagram showing dependencies and interfaces between modules.

Problem Analysis

The Smart Healthcare Portal consists of modules for patient management, appointment scheduling, medical records, billing, and notifications. The optimized Component Diagram defines the interactions and dependencies between these modules to ensure efficient and secure healthcare services.

Component Diagram



Explanation

- Patient Management handles patient registration and details.
- Appointment Scheduling manages booking and appointments.
- Medical Records stores patient health information.
- Billing processes payments.
- Notification Service sends reminders and confirmations.
- Authentication Service verifies user access.

Analysis

- Clearly defines component interactions.
- Shows module dependencies accurately.
- Improves modularity and maintainability.
- Ensures secure access through authentication.

Conclusion

The Component Diagram effectively represents the architecture of the Smart Healthcare Portal by showing all major components, interfaces, and dependencies, resulting in a secure and scalable system.

6. Deployment Diagram Design

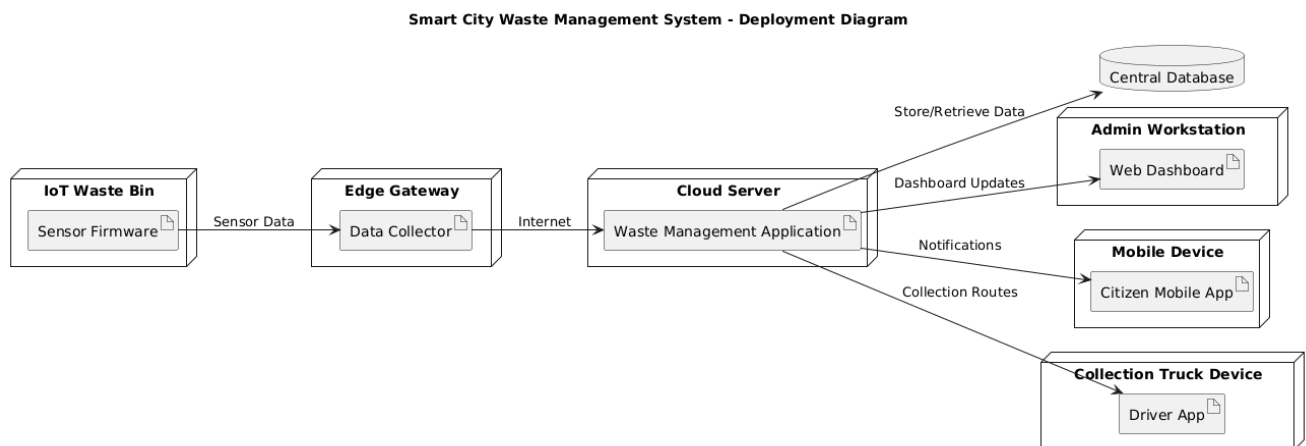
A Smart City Waste Management System is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.

Analyze the deployment environment, identify nodes, artifacts, and communication links, and construct a detailed Deployment Diagram representing the complete system architecture.

Problem Analysis

The Smart City Waste Management System uses IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards to monitor waste collection efficiently. The optimized Deployment Diagram shows the physical nodes, deployed artifacts, and communication links for reliable system operation.

Deployment Diagram



Explanation

- IoT Waste Bin Sensors collect bin-level data.
- Edge Device gathers and forwards sensor data.
- Cloud Server processes and stores information.
- Monitoring Dashboard displays real-time waste status.
- Mobile App sends alerts and notifications to users.

Analysis

- Clearly shows deployment nodes and communication links.
- Represents data flow from sensors to the cloud.
- Supports real-time monitoring and notifications.
- Provides a scalable and efficient deployment architecture.

Conclusion

The Deployment Diagram accurately represents the physical architecture of the Smart City Waste Management System by illustrating nodes, artifacts, and communication links, resulting in a clear and scalable deployment model.